

Task ID	Occupation Title	DWAs	Task Description
14675	Computer Systems Engineers/Architects	Monitor computer system performance to ensure proper operation.	Monitor system operation to detect potential problems.
18310	Acute Care Nurses	Operate diagnostic or therapeutic medical instruments or equipment. Prepare medical supplies or equipment for use.	Set up, operate, or monitor invasive equipment and devices, such as colostomy or tracheotomy equipment, mechanical ventilators, catheters, gastrointestinal tubes, and central lines.
4668.0	Gambling Cage Workers	Execute sales or other financial transactions.	Cash checks and process credit card advances for patrons.
15709	Online Merchants	Execute sales or other financial transactions.	Deliver e-mail confirmation of completed transactions and shipment.
6529	Kindergarten Teachers, Except Special Education	–	Involve parent volunteers and older students in children’s activities to facilitate involvement in focused, complex play.
6568	Elementary School Teachers, Except Special Education	–	Involve parent volunteers and older students in children’s activities to facilitate involvement in focused, complex play.

Table 1: Sample of occupations, tasks, and Detailed Work Activities from the O*NET database. We see that aggregating over activities alone is imprecise, as evidenced by the fact that we’d expect Gambling Cage Workers to complete the given DWA in person, using some physicality while we’d expect Online Merchants to complete the same activity solely with a computer.

DWA or complete a task by at least 50 percent. Though GPT-4 has vision capabilities OpenAI (2023b) and "LLM" is often used to refer to a much wider range of modalities, vision and image capabilities were only included in our definition of LLM-powered software. We provide a summary of our rubric below, while the complete rubric can be found in A.1. When we have labels for DWAs, we first aggregate them to the task level before aggregating to the occupation level.

Summary of exposure rubric

No exposure (E0) if:

- using the described LLM results in no or minimal reduction in the time required to complete the activity or task while maintaining equivalent quality^a or
- using the described LLM results in a decrease in the quality of the activity/task output.

Direct exposure (E1) if:

- using the described LLM via ChatGPT or the OpenAI playground can decrease the time required to complete the DWA or task by at least half (50%).

LLM+ Exposed (E2) if:

- access to the described LLM alone would not reduce the time required to complete the activity/task by at least half, but
- additional software could be developed on top of the LLM that could reduce the time it takes to complete the specific activity/task with quality by at least half. Among these systems, we count access to image generation systems.^b

^aEquivalent quality means that a third party, typically the recipient of the output, would not notice or care about LLM assistance.

^bIn practice, as can be seen in the full rubric in Appendix A.1, we categorize access to image capabilities separately (E3) to facilitate annotation, though we combine E2 and E3 for all analyses.

We set the exposure threshold at a potential 50% reduction in time required to complete a specific DWA

or task while maintaining consistent quality. We anticipate that adoption will be highest and most immediate for applications that realize a considerable increase in productivity. Although this threshold is somewhat arbitrary, it was selected for ease of interpretation by annotators. Moreover, regardless of the chosen threshold, we guessed that the real-world reduction in task time would likely be slightly or significantly lower than our estimates, leading us to opt for a relatively high threshold. In our own validation labeling, we found that this corresponded closely to whether an LLM or LLM-powered software could perform the core part of a task or nearly the entire task.

Comparison	γ	Weighting	Agreement	Pearson's
GPT-4, Rubric 1; Human	α	E1	80.8%	0.223
	β	E1 + .5*E2	65.6%	0.591
	ζ	E1 + E2	82.1%	0.654
GPT-4, Rubric 2; Human	α	E1	81.8%	0.221
	β	E1 + .5*E2	65.6%	0.538
	ζ	E1 + E2	79.5%	0.589
GPT-4, Rubric 1; GPT-4, Rubric 2	α	E1	91.1%	0.611
	β	E1 + .5*E2	76.0%	0.705
	ζ	E1 + E2	82.4%	0.680

Table 2: Model and human comparison of agreement and Pearson's correlation scores. The agreement score is determined by looking at how often the two groups agree on the annotation (e.g. E0, E1 or E2). In the paper we use GPT-4, Rubric 1. Core tasks are given twice the weight at the occupation-level as supplemental tasks. All weights sum to one.

We then collected both human and GPT-4-generated annotations using the exposure rubric, which underlie the bulk of the analyses in this paper.

- *Human Ratings:* We obtained human annotations by applying the rubric to each O*NET Detailed Worker Activity (DWA) and a subset of all O*NET tasks and then aggregated those DWA and task scores⁵ at the task and occupation levels. The authors personally labeled a large sample of tasks and DWAs and enlisted experienced human annotators who have reviewed GPT-3, GPT-3.5 and GPT-4 outputs as part of OpenAI's alignment work (Ouyang et al., 2022).
- *GPT-4 Ratings:* We administered a similar rubric to an early version of GPT-4 (OpenAI, 2023b) but on all task/occupation pairs rather than DWAs. We made slight modifications to the rubric (which was used as a "prompt" to the model in this case) to enhance agreement with a set of human labels. Full agreement rates are given in Table 2.

We construct three primary measures for our dependent variable of interest: (i) α , corresponding to E1 in the exposure rubric above, anticipated to represent the lower bound of the proportion of exposed tasks within an occupation, (ii) β , which is the sum of E1 and 0.5*E2, where the 0.5 weight on E2 is intended to account for exposure when deploying the technology via complementary tools and applications necessitates additional investment, and (iii) ζ , the sum of E1 and E2, an upper bound of exposure that provides an assessment of maximal exposure to an LLM and LLM-powered software. We summarize agreement between annotation groups and measures in Table 2. For the remainder of the analysis, if not specified, the reader may assume that

⁵The authors annotated DWAs that clearly required a high degree of physicality or manual dexterity, and the contracted annotators labeled the remaining activities, along with a subset of tasks including those without associated DWAs and those for which there was no clear task-level annotation after aggregating the DWA annotations.